

# Monitoring and Logging Plan

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Project: Implementing a Secure Network Architecture for a Large Enterprise

Organisation: Spinnaker Analytics

## 1. Purpose and Scope

- This document defines the network monitoring and centralised logging strategy for GlobalTech Corp.
- It remediates Finding 14 from the Network Architecture Assessment Report (22 May 2026):
  - Finding 14: No centralised logging or SIEM
    - security logs siloed per device with no correlation capability, severely hampering incident detection and forensic investigation
- This plan covers log collection from all network zones, devices, and remote sites; centralised aggregation in a SIEM platform;
- alerting rules for suspicious activities; and dashboards for continuous monitoring.
- All log sources documented in previous deliverables feed into this centralised platform.

## 2. SIEM Platform

### 2.1 Solution Selection

- Elastic SIEM (Elastic Security) is selected as the centralised SIEM platform, deployed on the Server VLAN at HQ.
- The Elastic Stack (Elasticsearch, Logstash, Kibana) provides log ingestion, storage, correlation, alerting, and visualisation in a single platform:

| Component     | Role                                   | Deployment                                  |
|---------------|----------------------------------------|---------------------------------------------|
| Elasticsearch | Log storage and search engine          | 3-node cluster — 10.10.20.60/61/62          |
| Logstash      | Log ingestion and parsing pipeline     | 10.10.20.63 — receives from all sources     |
| Kibana        | Dashboards, alerting, investigation UI | 10.10.20.64 — HTTPS only, MFA-gated         |
| Elastic Agent | Unified log shipper on endpoints       | Deployed on all managed endpoints via Fleet |
| Fleet Server  | Manages Elastic Agent deployments      | 10.10.20.65 — central agent management      |

### 2.2 Log Retention Policy

| Log type                        | Retention period | Storage tier                          |
|---------------------------------|------------------|---------------------------------------|
| Security alerts and SIEM events | 2 years          | Hot tier — fast search                |
| Firewall and network logs       | 1 year           | Warm tier after 90 days               |
| Authentication and IAM logs     | 2 years          | Hot tier for 90 days, warm thereafter |
| Endpoint EDR events             | 1 year           | Warm tier after 30 days               |
| Application logs                | 90 days          | Hot tier                              |
| VPN access logs                 | 2 years          | Warm tier after 90 days               |
| Raw syslog (all other)          | 90 days          | Cold tier after 30 days               |

## 3. Log Sources

### 3.1 Log Source Inventory

- The following log sources are configured to forward to the Elastic SIEM:

| Source                   | Log type                                        | Transport                               | Format            |
|--------------------------|-------------------------------------------------|-----------------------------------------|-------------------|
| NGFW (pfSense/FortiGate) | Firewall allow/deny, app-layer events           | Syslog TLS → Logstash                   | CEF / syslog      |
| Suricata IDPS            | IDS/IPS alerts, dropped packets                 | Filebeat → Elasticsearch                | eve.json (JSON)   |
| Core L3 switch           | Interface events, ACL hits, MAC changes         | Syslog UDP → Logstash                   | Syslog            |
| VPN gateway (WireGuard)  | Connection events, auth failures, tunnel status | Filebeat → Elasticsearch                | JSON              |
| Wazuh EDR agents         | Process events, FIM, vulnerability scan, alerts | Wazuh → Elasticsearch                   | JSON              |
| Microsoft Entra ID       | Sign-in logs, MFA events, PIM activations       | Entra diagnostic → Event Hub → Logstash | JSON              |
| Active Directory         | Logon events, group changes, policy changes     | Winlogbeat → Elasticsearch              | Windows Event Log |
| Windows endpoints        | Security, application, system event logs        | Elastic Agent → Fleet                   | Windows Event Log |
| Linux servers            | Auth, syslog, audit logs                        | Elastic Agent → Fleet                   | Syslog / auditd   |
| HashiCorp Vault          | Key access, policy changes, auth events         | Vault audit log → Filebeat              | JSON              |
| Remote office firewalls  | Firewall events (Delhi, Bangalore)              | Syslog TLS → Logstash (via VPN)         | CEF / syslog      |
| Web proxy (Squid)        | HTTP/HTTPS request logs, blocked URLs           | Filebeat → Elasticsearch                | Squid native log  |

## 3.2 Logstash Pipeline Configuration

- Logstash processes incoming logs, normalises them to ECS (Elastic Common Schema), enriches with GeoIP and threat intelligence, and forwards to Elasticsearch:

```
# Logstash pipeline - firewall syslog input
input {
  syslog {
    port => 6514
    ssl_enable => true
    ssl_cert => '/etc/logstash/certs/logstash.crt'
    ssl_key => '/etc/logstash/certs/logstash.key'
    type => 'firewall'
  }
}

filter {
  if [type] == 'firewall' {
    grok { match => { 'message' => '%{SYSLOGLINE}' } }
    geoip { source => 'src_ip' target => 'source.geo' }
    translate {
      field => 'src_ip'
      destination => 'threat.indicator'
      dictionary_path => '/etc/logstash/threat-intel.yml'
    }
  }
}

output {
  elasticsearch {
    hosts => ['10.10.20.60:9200','10.10.20.61:9200']
    index => 'logs-firewall-%{+YYYY.MM.dd}'
    ssl => true
    cacert => '/etc/logstash/certs/ca.crt'
  }
}
```

## 4. Alerting Rules

### 4.1 Critical Alerts — Immediate Response Required

| Alert name               | Trigger condition                                                      | Response                                             |
|--------------------------|------------------------------------------------------------------------|------------------------------------------------------|
| Ransomware behaviour     | Mass file modifications + shadow copy deletion on same host within 60s | Auto-isolate endpoint via Wazuh active response      |
| C2 beaconing detected    | IDPS ET TROJAN rule triggered — outbound to known C2 IP                | Block IP at NGFW, isolate source host, alert SOC     |
| Admin account compromise | Admin sign-in from new country or impossible travel detected           | Suspend account, require re-verification, alert CISO |
| Brute force — VPN        | 10+ FAILED_AUTH from same IP within 2 minutes                          | Block IP at VPN gateway, alert security team         |
| Lateral movement         | PsExec or WMI remote execution detected on internal hosts              | Isolate source and destination, alert SOC            |
| Data exfiltration        | Outbound transfer >100MB to external IP outside business hours         | Block connection, alert security team, preserve logs |
| Privilege escalation     | User account added to admin group outside change window                | Revert change, alert security team, review account   |

### 4.2 High Priority Alerts — Investigate Within 1 Hour

| Alert name                  | Trigger condition                                      | Response                                                     |
|-----------------------------|--------------------------------------------------------|--------------------------------------------------------------|
| Port scan detected          | 15+ SYN packets from same src IP within 3 seconds      | Log source IP, check for follow-up exploitation attempts     |
| SQL injection attempt       | WAF rule 942100 triggered on web app                   | Review request, check for successful bypass, patch if needed |
| MFA fatigue attack          | User receiving 10+ MFA push requests within 5 minutes  | Temporarily block MFA push, notify user, investigate source  |
| Unpatched device on network | NAC detects endpoint >30 days behind on patches        | Quarantine to remediation VLAN, notify user and IT team      |
| Failed Vault access         | 5+ consecutive failed auth attempts to HashiCorp Vault | Lock out source, review Vault audit log, alert security      |
| Certificate expiry warning  | TLS certificate expiring within 14 days                | Trigger renewal workflow, escalate if renewal fails          |

## 4.3 Elastic SIEM Detection Rule — Example

- The following is a sample Elastic SIEM detection rule for brute-force VPN authentication:

```
# Elastic SIEM - VPN brute force detection rule
name: VPN Brute Force - 10+ failures in 2 minutes
type: threshold
index: logs-vpn-*
query: event.action:FAILED_AUTH AND destination.port:443
threshold:
  field: source.ip
  value: 10
  cardinality:
    - field: user.name
      value: 3
window: 2m
severity: high
actions:
  - type: webhook
    url: https://soc.globaltech.internal/alert
  - type: email
    to: security@globaltech.com
```

## 5. Dashboards

### 5.1 Security Operations Dashboard

- The primary SOC dashboard provides a real-time overview of security events across all zones. Key panels:
  - Alert timeline
    - all SIEM alerts by severity over last 24 hours
  - Top blocked source IPs
    - NGFW deny events ranked by frequency
  - Authentication summary
    - successful vs failed logins by source
    - MFA failures
  - IDPS alert breakdown
    - top triggered Suricata rule categories
  - Endpoint health map
    - Wazuh agent status across all sites
  - Active VPN sessions
    - current connected users and session durations

### 5.2 Network Traffic Dashboard

- Traffic volume by VLAN
  - bandwidth per segment over time
- Top talkers
  - source/destination pairs by volume
- Protocol distribution
  - breakdown of traffic by protocol across each zone
- Geo map
  - inbound connections by country of origin
- DMZ traffic
  - all traffic entering and leaving the DMZ

## 5.3 Compliance Dashboard

- Patch compliance
  - percentage of endpoints within 30-day patch window
- Encryption status
  - endpoints with disk encryption enabled vs non-compliant
- MFA registration
  - accounts with MFA registered vs pending
- Certificate expiry
  - all TLS certificates with days to expiry
- Access review status
  - open vs completed quarterly review tasks



## 6. Network Monitoring

### 6.1 Continuous Network Monitoring Tools

| Tool            | Purpose                                              | Deployment                        |
|-----------------|------------------------------------------------------|-----------------------------------|
| Elastic SIEM    | Central correlation, alerting, investigation         | HQ Server VLAN                    |
| Suricata        | Network IDS/IPS — traffic inspection                 | Inline on NGFW WAN interface      |
| Wazuh           | Endpoint monitoring, FIM, vulnerability scan         | Agent on all endpoints            |
| ntopng          | Network flow analysis, bandwidth monitoring          | Core switch SPAN port             |
| Nagios / Zabbix | Infrastructure health — uptime, CPU, memory          | HQ Server VLAN                    |
| Canary tokens   | Honeypot files/credentials — detect internal threats | Planted in Finance/HR file shares |

### 6.2 Canary Token Implementation

- Canary tokens are decoy files and credentials planted in sensitive directories.
- Any access to these files immediately alerts the security team, indicating either an external attacker or a malicious insider has accessed restricted areas:
  - Finance directory:
    - fake spreadsheet named 'Salary\_2026\_Confidential.xlsx'
    - access triggers immediate alert
  - HR directory:
    - fake employee database export
    - access triggers immediate alert
  - Domain Admin credentials:
    - canary credential set planted in password manager
    - any use triggers lockdown
  - Alerts delivered via:
    - SIEM alert + email to CISO + Slack message to security channel

## 7. Testing and Verification

| Test                        | Method                                             | Expected result                                      |
|-----------------------------|----------------------------------------------------|------------------------------------------------------|
| Log ingestion — all sources | Verify each source appears in Kibana Discover      | All 12 log sources visible with recent events        |
| Alert — VPN brute force     | Run script generating 15 failed VPN auths in 2 min | SIEM alert triggered, email sent to security team    |
| Alert — C2 beaconing        | Curl to known C2 IP from test endpoint             | Suricata alert in SIEM, NGFW block rule triggered    |
| Alert — after-hours admin   | Sign in as admin at 02:00 IST                      | Step-up MFA required, SIEM alert generated           |
| Canary token — Finance file | Open canary file from test account                 | Alert within 60 seconds to SIEM and email            |
| Dashboard accuracy          | Compare dashboard counts to raw log queries        | Counts match within 1% — no data loss in pipeline    |
| Log retention               | Query for 90-day-old logs                          | Logs accessible, correctly tiered to warm storage    |
| Remote site log forwarding  | Check Delhi and Bangalore logs in SIEM             | Both sites forwarding logs via VPN tunnel to HQ SIEM |

## 8. Maintenance Schedule

| Task                                                           | Frequency           |
|----------------------------------------------------------------|---------------------|
| Review SIEM alerts — triage and close or escalate              | Daily               |
| Check log pipeline health — verify all sources ingesting       | Daily               |
| Review and tune alert rules — reduce false positives           | Weekly              |
| Update Suricata rulesets (ET Open)                             | Daily (automated)   |
| Review Elasticsearch cluster health and storage usage          | Weekly              |
| Test all critical alert rules with simulated events            | Monthly             |
| Archive logs to cold storage per retention policy              | Monthly (automated) |
| Full SIEM health check — performance, storage, ingestion rates | Quarterly           |

## 9. Summary

- The monitoring and logging plan establishes complete visibility across all GlobalTech Corp systems and sites. Controls now in place:
  - Elastic SIEM deployed
    - 3-node Elasticsearch cluster
    - Logstash ingestion pipeline
    - Kibana dashboards
  - 12 log sources configured
    - NGFW
    - IDPS
    - Switch
    - VPN
    - EDR
    - IAM
    - AD
    - Endpoints
    - Vault
    - remote offices
    - proxy
  - 7 critical alert rules
    - Ransomware
    - C2
    - admin compromise
    - brute force
    - lateral movement
    - Exfiltration
    - privilege escalation
  - 6 high-priority alert rules
    - port scan
    - SQLi
    - MFA fatigue
    - unpatched device
    - Vault failure
    - certificate expiry
  - 3 Kibana dashboards
    - security operations
    - network traffic
    - compliance
  - Canary tokens deployed in Finance and HR directories
    - insider threat detection
  - Log retention policy
    - 2 years for security events
    - tiered storage for cost efficiency
  - All 8 test scenarios verified — log ingestion, alert firing, canary token, remote site forwarding